

	MPS Engineering Work Order for In-kind Replacement (Non MOC) Инженерная Инструкция для Работ MPS по Аналогичной Замене (Без КЗИ)								
	Anomaly Report: Отчет об аномалии:	N/A	EWR #: # ЗИР:	25-0073	Subsystem: Подсистема:	TL	Line №: № Линии:	63-9100-TL-1085-6"-150H22-HCW5	
	Area: Участок:	SGP	WO: РЗ:	1799855	P&ID #: СТИКИП	62-0200-B-PID-0201-02	Title: Название:	TA26 TLR# 23154 REMOVAL ON TL LINE	
TENGIZCHEVROIL ТЕНГИЗШЕВРОЙЛ	Unit: Установка:	U9100	Drawing/Sketch No: Чертеж/Схема No:	063-9100-LLL-ISO-20068-01					
Standard of Acceptance: Стандарт Приемки:	B31.3		TCO Specification: Спецификации ТШО:	W-ST-2025			TCO Approved WPS: Утвержденная Сварочная Процедура ТШО:	20AW/ 21AW	
Is wall thickness measurement required at tie-in points prior to start works (to verify existing pipe sound wall thickness) Требуется ли замер толщины стенки на точках врезки перед началом работ (для определения приемлемой толщины существующего трубопровода)			YES		Is 100% PT/MT required for weld bevel ends prior to welding Требуется ли 100% ККД/МПД для проверки кромок перед сваркой			YES/ДА	
Shop Hydro-test Испытание Гидро-тестом в Цеху	NA/НП barg		Piping NDT НК Трубопровода	NA/НП VT/BK 100%	Coating System #: № Системы Покрытия:	12.1	Hydrogen Bake out Водородный Отжиг	NA/НП	
Field Hydro-test Испытание Гидро-тестом на Участке	NA/НП barg		Piping Support Structural NDT НК Опорной Конструкции Трубопровода	NA/НП	Insulation type and thickness: Тип и толщина изоляции:	HCW5 60mm	PMI for Alloy materials (SP 20) ХАМ для Материалов из сплавов (ПТБ 20)	NA/НП	
HARDNESS TESTING (Maximum up to 200HB) ОПРЕДЕЛЕНИЕ ТВЕРДОСТИ (Максимум до 200НВ)	NA/НП		Stress Relief (PWHT) Снятие Напряжений (ПСТО)	NA/НП	Notes Примечания				

Job Description
Описание работ

see next page for detailed instruction. ***** Смотреть следующую страницу для детальной информации.

Approved by Потверждено	Name Имя	Signature Подпись
Originated by: Подготовлено:	DE MPS Engineer Инженер DE MPS	Adil Maxutov Digitally signed by Adil Maxutov Date: 2025.03.01 12:38:43 +05'00'
Approved by: Утверждено:	DE Engineer-Consultant DE Инженер-Консультант проектной группы	Bekzat Abzaliyev Digitally signed by Bekzat Abzaliyev Date: 2025.03.17 15:53:53 +05'00'
Reviewed by: Рассмотрено:	RE QM Lead: Вед. Специалист КК Отдела Надежности:	
Reviewed by: Рассмотрено:	FER Authorized Inspector: Уполномоченный инспектор НСО:	
Reviewed by: Рассмотрено:	TA Group Supervisor: Руководитель группы КР:	
Reviewed by: Рассмотрено:	TA Unit Coordinator: Координатор КР установок ОЗ:	
Approved by: Утверждено:	Design Engineering Supervisor: Руководитель инженерно-проектной группы:	

PRESENT SITUATION:

- There is leak in the pipe section 63-9100-TL-1085-6"-150H22-HCW5. TLR was installed TAG 2 23154.

PROPOSED SOLUTION:

- Replace damaged section of 63-9100-TL-1085-6"-150H22-HCW5 line due to leak.

SCOPE OF WORK

- Contractor shall strictly follow all relevant TCO Safety Instructions in the progress of works, all of which are readily obtainable from the TCO Intranet.
- Contractor shall follow all Quality Management (QM) requirements designated in the Quality Control package.

SHOP WORK:

- Withdraw materials required for this Job Pack from TCO Warehouse according to Material Request(s).
- Prefabricate pipe spools, as per isometric drawing(s) 063-9100-LLL-ISO-20068-01 and 063-9100-LLL-ISO-20069-01. Use TCO-approved Welding Procedure Specification(s) (WPSs) indicated on isometric drawing.
- Perform non-destructive examination (NDE) of shop welds, as specified on isometric drawing 63-9100-LLL-ISO-20068-01 and 063-9100-LLL-ISO-20069-01.

SITE WORK:

- Perform relevant job safety assessments (PPHA & JSA) and obtain required Permits-to-Work (PTWs).
- Demarcate work area as necessary.

NOTE: TCO Operations shall isolate, depressurize, drain and ready for Hot Work, Equipment and Piping associated with this Project, if and as applicable, and shall hand it over to CONTRACTOR; fully accessible, safe and Locked-Out-Tagged-Out (LOTO) as scheduled and agreed between TCO Operations and CONTRACTOR.

- Remove all cladding and insulation to the extent required.
- Arrange with Zone Maintenance to isolate, LOTO and remove electric trace heating and instruments from associated piping.

NOTE: CONTRACTOR to match mark and record location prior to cutting and removal to aid in reinstatement of heat tracing.

- Cut, unbolt and remove existing piping shown on "Destruct Detail" of isometric drawing(s) 063-9100-LLL-ISO-20068-01.

- Install piping as specified on isometric drawing 063-9100-LLL-ISO-20068-01 and 063-9100-LLL-ISO-20069-01. For field welds use TCO-approved WPSs indicated on isometric drawing(s).
- Perform NDE of field welds, as specified on isometric drawing 63-9100-LLL-ISO-20068-01 and 063-9100-LLL-ISO-20069-01.
- Perform surface preparation and touch up field welds and any other coating damages in accordance with TES COM-SU-4743-TCO and coating system(s) per TES COM-SU-5191-TCO specified on drawings.
- Arrange with Zone Maintenance to reinstate / install related electric trace heating and instruments.
- All QM piping passport documentation (A-category Checklists) to be verified and signed by TCO-approved QM Inspector prior to PSSR, but not later than 24 hours after mechanical scope completion.
- Advise Mursal Rauan / Ramazanov Nurbulat DE TA26 SGP/SGI Mechanical Engineers at ext. 4259 about work completion and invite all necessary people to PSSR.
- Job Pack Coordinator shall submit the completed PIC check-list to Operations department after being signed by Quality Management group. (Form is provided by QM Dept.).
- Conduct PSSR and close out A-category punch points.
- Following PSSR reinstate / install insulation and cladding, as per TES IRM-SU-1381-TCO and as specified on isometric drawing.
- Close out all B-category punch points and remove associated scaffolding.
- Clean up the work area and demobilize from site.

AFTER COMPLETION OF WORK:

- Perform service test.
- Final as-built package (including B-category Checklist) shall be submitted to TCO QM group within 2 weeks after completion of the whole scope of works.
- Return excess material to TCO Warehouse.

REFERENCED TCO SPECS:

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|-------------------|------------------------------------|
| • A-ST-2008 | Basic Engineering Data |
| • SI-105 | Permit to work and hazard analysis |
| • W-ST-2025 | Welding, PWHT and NDE of piping |
| • PIM-SU-2505-TCO | Carbon Steel Piping Fabrication |
| • RM-SU-1381-TCO | Thermal Insulation for Hot Lines |